Towards Type Systems That Reinterpret Themselves

(supplementary materials)

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1 A type of internal models of MLTT-R

A comprehension $[x : A, y : B, z : C]$ denotes a right-nested Σ-type. $\{x : A\} \to B$ is a syntactic sugar meaning that argument x is always inferred from B rather than provided explicitly. $\{x\} \to B$ means that type of x is inferred from B. Transports along previously introduced equalities are omitted. Since our language doesn't yet have η-equivalence, uniqueness properties are skipped.

$$\mathtt{modelDef}(\ell_0, \ldots, \ell_\kappa) = [$$

1.1 Base CwF part

$$
\begin{aligned}
&C : \mathcal{U}_{\max(\ell_0,\ldots,\ell_\kappa)+1},\\
&\triangleright : C \to C \to \mathcal{U}_{\max(\ell_0,\ldots,\ell_\kappa)},\\
&\mathrm{id} : \{\Gamma : C\} \to \Gamma \triangleright \Gamma,\\
&\circ : \{\Gamma\ \Delta\ \Xi\} \to (\Delta \triangleright \Xi) \to (\Gamma \triangleright \Delta) \to (\Gamma \triangleright \Xi),\\
&\mathrm{id}{-}\circ : \{\Gamma, \Delta\} \to \{\sigma : \Gamma \to \Delta\} \to \mathrm{id} \circ \sigma = \sigma,\\
&\circ{-}\mathrm{id} : \{\Gamma, \Delta\} \to \{\sigma : \Gamma \to \Delta\} \to \sigma \circ \mathrm{id} = \sigma,\\
&\circ{-}\mathrm{assoc} : \{\rho\ \sigma\ \tau\} \to \rho \circ (\sigma \circ \tau) = (\rho \circ \sigma) \circ \tau,\\
&\cdot : C,\\
&! : \{\Gamma\} \to \Gamma \triangleright \cdot,
\end{aligned}
$$

1.2 Common part for each level j

$$
\begin{aligned}
&\mathrm{Ty}_j : C \to \mathcal{U}_{\ell_j+1},\\
&_[_] : \{\Gamma\ \Delta\} \to \mathrm{Ty}_j \Gamma \to \Delta \triangleright \Gamma \to \mathrm{Ty}_j \Delta,\\
&\mathrm{sub/id} : \{T\} \to T[\mathrm{id}] = T,\\
&\mathrm{sub/comp} : \{T\ \sigma\ \tau\} \to T[\sigma \circ \tau] = T[\sigma][\tau],\\
&\mathrm{Tm}_j : (\Gamma : C) \to \mathrm{Ty}_j\ \Gamma \to \mathcal{U}_{\ell_j},\\
&_[_] : \{\Gamma\ \Delta\ T\} \to \mathrm{Tm}_j\ \Gamma\ T \to (\sigma : \Delta \triangleright \Gamma) \to \mathrm{Tm}_j\ \Delta\ T[\sigma],\\
&\mathrm{sub/id} : \{T\} \to \{t : \mathrm{Tm_}\ T\} \to t[\mathrm{id}] = t,\\
&\mathrm{sub/comp} : \{T\ \sigma\ \tau\} \to \{t : \mathrm{Tm}_j\ _\ T\} \to t[\sigma \circ \tau] = t[\sigma][\tau],\\
&(.) : (\Gamma : C) \to \mathrm{Ty}_j\ \Gamma \to C,\\
&p : \{\Gamma\ T\} \to \Gamma.T \triangleright \Gamma,\\
&q : \mathrm{Tm}\ \Gamma.T\ T[p],\\
&\langle , \rangle : \{\Delta\ \Gamma\} \to (\sigma : \Delta \triangleright \Gamma) \to \mathrm{Tm}\ \Delta\ T[\sigma] \to \Delta \triangleright \Gamma.T,\\
&\mathrm{ext/l} : \{\sigma\ t\} \to p \circ \langle \sigma, t \rangle = \sigma,\\
&\mathrm{ext/r} : \{\sigma\ t\} \to q[\langle \gamma, t \rangle] = t,\\
&\mathrm{Nat}_j : \{\Gamma\} \to \mathrm{Ty}_j\ \Gamma,\\
&0 : \{\Gamma\} \to \mathrm{Tm}_j\ \Gamma\ \mathrm{Nat}_j,\\
&S : \{\Gamma\} \to \mathrm{Tm}_j\ \Gamma\ \mathrm{Nat}_j \to \mathrm{Tm}_j\ \Gamma\ \mathrm{Nat}_j,\\
&\mathrm{Id} : \{\Gamma\} \to (T : \mathrm{Ty}_j\ \Gamma) \to \mathrm{Tm}_j\ \Gamma\ T \to \mathrm{Tm}_j\ \Gamma\ T \to \mathrm{Ty},\\
&\mathrm{refl} : \{\Gamma\ T\} \to \{t\ t' : \mathrm{Tm}_j\ \Gamma\ T\} \to t = t' \to \mathrm{Tm}_j\ \Gamma\ (\mathrm{Id}\ T\ t\ t'),
\end{aligned}
$$

1.3 Common part for each j less then k

Here, definition of a model contains internalization of universes and of type family El at internal level $j + 1$.

1.4 Common part for each i, j

Here, definition of a model contains eliminators for model of naturals and for model of an identity type; additionally, models of Π- and Σ- types for internal level $\max(i, j)$ are defined here.

1.5 End of definition of a model type

]

2 Parts of a definition of a displacement model

$$
\begin{aligned}
\uparrow_\ell^\kappa = \ & [\\
& C = \mathcal{U}_{\ell+\kappa}, \triangleright = (\to), \mathrm{id} = \lambda x.x, \circ = \lambda f g x.f(gx), \mathrm{id}{-}\circ = \mathrm{refl}, \circ{-}\mathrm{id} = \mathrm{refl}, \\
& \ldots, \\
& \mathrm{Ty}_j = \lambda \Gamma.\Gamma \to \mathcal{U}_{j+\kappa}, \mathrm{Tm}_j = \lambda \Gamma\, f.(\gamma : \Gamma) \to f\,\gamma, \\
& \ldots \\
&]
\end{aligned}
$$

3 Parts of a definition of a parametricity model

$$
\begin{aligned}
\mathrm{Para}_\ell = \ & [\\
& C = (\Gamma\,\Gamma' : \mathcal{U}_{\ell+1}) \times (\Gamma \to \Gamma' \to \mathcal{U}_{\ell+1}), \\
& \triangleright = \lambda(\Gamma, \Gamma', Q)(\Delta, \Delta', R).(f : \Gamma \to \Delta) \times (f' : \Gamma' \to \Delta') \times ((\gamma\,\gamma') \to Q\,\gamma\,\gamma' \to R\,(f\,\gamma)\,(f'\,\gamma')), \\
& \mathrm{id} = (\lambda x.x, \lambda x.x, \lambda x.x), \\
& \circ = \lambda(f, f', q)(g, g', r)(x, x', s).(f(g(x)), f'(g'(x')), q(r(s))), \\
& \mathrm{id}{-}\circ = \mathrm{refl}, \circ{-}\mathrm{id} = \mathrm{refl}, \\
& \ldots, \\
&]
\end{aligned}
$$